PROJECT CHECKPOINT — WEEK 3 OF 8

Predicting Demand & Detecting Anomalies in Urban Bike-Share Systems

Project Checkpoint — Progress Update

Hoa (thaycacac) • Data Mining Project Course • October 2026

AGENDA

Outline

- Recap: Problem, Related Work & Approach
- Progress Update: What's Done So Far
- Preliminary Results
- Evaluation Plan
- Updated Timeline & Next Steps

RECAP

Problem & Motivation

- Bike-share stations suffer demand-supply imbalance across the day
- Manual rebalancing is costly and reactive
- Anomalous trips (theft, GPS errors, malfunction) go undetected without systematic monitoring

> **Research question:** Can trip, weather, and calendar data predict near-term station demand and flag anomalous trips worth investigating?

RECAP

Approach: Data, Tools & Techniques

Stage	Technique	Purpose
Preprocessing / Warehousing	Cleaning + star schema	Reliable, query-ready trip data
Clustering	k-means / DBSCAN	Station usage-profile groups
Classification	Random Forest / GBM	Predict high/low demand station-hours
Frequent patterns	FP-Growth	Common origin-destination sequences
Outlier detection	Isolation Forest	Flag anomalous trips

Data: Citi Bike NYC trips + NOAA weather + calendar | Tools: pandas, scikit-learn, mlxtend, PyOD, SQLite

PROGRESS UPDATE

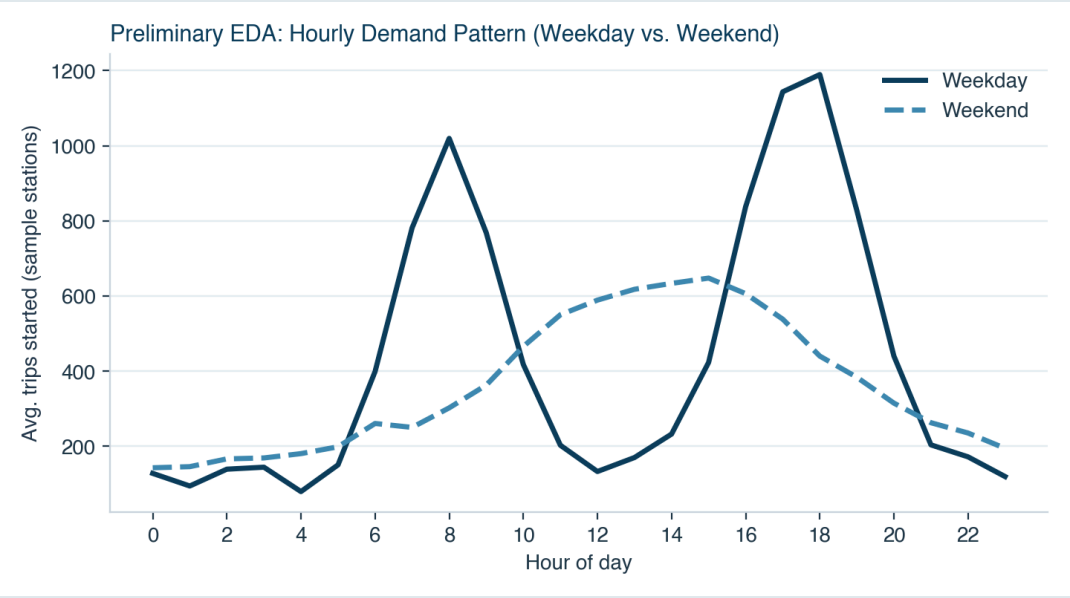
What's Done So Far (Weeks 1–3)

- **Data acquired & cleaned:** 3 months of Citi Bike trips (~1.4M after cleaning), merged with weather + calendar data
- **Data warehouse built:** star-schema in SQLite (trip fact + station/time/weather dimensions)
- **Exploratory analysis complete:** clear weekday vs. weekend demand patterns confirmed
- **Clustering complete:** k=4 station usage-profile clusters, silhouette ≈ 0.42

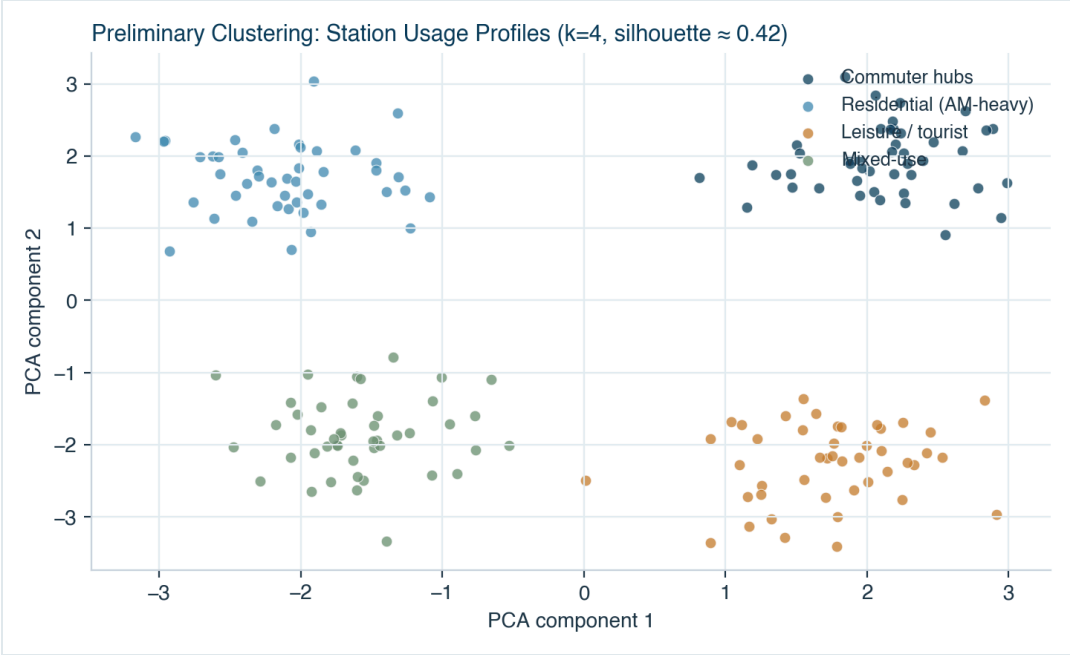
> **Feedback incorporated:** scope narrowed to Manhattan stations (per peer feedback on tractability); classifier baseline clarified as historical station-hour-of-week average.

PRELIMINARY RESULTS

EDA & Clustering Findings



Weekday vs. weekend hourly demand



Station usage-profile clusters (k=4)

PROGRESS UPDATE

Challenges & Adjustments

- Weather feed had small gaps — filled via linear interpolation
- Station usage vectors were skewed by a few high-traffic stations — log-scaling improved cluster separation
- ~3% of raw trip records dropped due to invalid station IDs / implausible timestamps

EVALUATION PLAN

How We'll Measure Success

- **Classification:** F1-score vs. historical-average baseline (target: +10%)
- **Clustering:** Silhouette score (achieved: ≈0.42) + interpretability
- **Frequent patterns:** Support / confidence / lift, manually reviewed for plausibility
- **Outlier detection:** Manual review precision on flagged trip sample

UPDATED TIMELINE

Project Phases & Status

Phase	Weeks	Status
Data acquisition & preprocessing	1–2	✓ Completed
Exploratory analysis & clustering	3	✓ Completed
Classification & pattern mining	4–5	• In Progress
Outlier detection & evaluation	6	Upcoming
Final report & presentation	7–8	Upcoming

Next Steps

- Train & tune demand classifier (Weeks 4–5)
- Mine frequent origin-destination patterns (Weeks 4–5)
- Run outlier detection & manual review (Week 6)
- Final evaluation, report & presentation (Weeks 7–8)

Questions & Discussion — Thank you!